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7. Three different radioactive isotopes, **X**, **Y** and **Z** are investigated.
The table below shows the count rate detected when different absorbers are placed between each radioactive isotope and a detector.
The distance between the detector and the radioactive isotopes is fixed.

The count rates have been corrected for background radiation.

Radioactive isotope	Count rate (units)			
	No absorber	Paper	Aluminium	Lead
X	20	20	20	6
Y	74	74	56	15
Z	44	32	32	12

(a) Explain how to correct count rates for background radiation. [2]

....

....

....

(b) The radioactive isotopes used and the radiation they emit is shown in the table below.

Radioactive isotope	Radiation emitted
silver-110	beta and gamma
cobalt-60	gamma
radium-226	alpha and gamma

Use the information in both tables to identify the radiation(s) emitted by **X and Y** and identify these isotopes.

(i) Radioactive isotope **X** emits ....

Radioactive isotope **X** is .... [2]

(ii) Radioactive isotope **Y** emits ....

Radioactive isotope **Y** is .... [3]

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(c) One isotope of silver has the symbol $^{110}_{47}\mathrm{Ag}$.

Complete the following sentences using numbers from the box.

47	63	110	157

(i) The number of protons in an atom of silver is .... [1]

(ii) The number of neutrons in an atom of this isotope of silver is .... [1]

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END OF PAPER